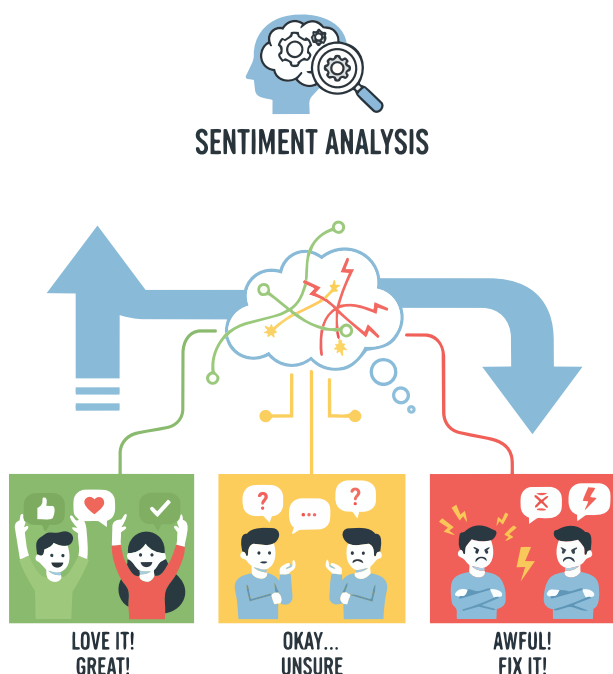


BERT: The Aspect-Based Sentiment Analyser

While BERT is great for sentiment analysis in natural language processing, it has a few limitations too.



Sentiment analysis of a complex sentence often requires a contextual understanding of the entire sentence rather than understanding the words in isolation or from left-to-right. Such contextual understanding requires simultaneous bi-directional reading of the sentence. Bidirectional Encoder Representations from Transformers, i.e., BERT, is considered the gold standard for Aspect-Based Sentiment Analysis (ABSA) because of its unique ability to understand context and directionality of a sentence and has become non-negotiable for it.

ABSA is an advanced NLP technique that can analyse texts beyond general sentiment (e.g., “This phone is okay”) to specific feature-level insights (e.g., “The battery is great (+), but the camera is blurry (-)”).

The ABSA concept can be divided into four key terms.

Aspect Terms (AT): The specific words used to describe a feature (e.g., ‘attended’, ‘cost’).

Aspect Categories (AC): The categories of the aspect term (e.g., ‘Service’, ‘Hardware’).

Opinion Terms (OT): The adjectives used to qualify the aspect term (e.g., ‘slow’, ‘brilliant’).

Sentiment polarity: The final classification—Positive, Negative, or Neutral.

Implementation pipeline

Step 1: Aspect Extraction (AE)

Identify the aspect term. For this, rule-based or supervised methods can be used. In the rule-based approach, POS (parts of speech) tagging is used to find noun or noun phrases as aspects. Secondly, in a supervised method, a BERT-based named entity recognition (NER) model trained to label aspect terms is used.

Step 2: Sentiment Classification (SC)

To identify the sentiment with respect to the aspect term, feed the model with both the sentence and the aspect as a pair: [CLS] Sentence [SEP] Aspect [SEP].

Step 3: Triplet Extraction (ASTE)

Extract sentiment as a triplet: (Aspect, Sentiment, Opinion); for example: (‘mobile’, ‘positive’, ‘good’).

To train a model, the following industry-standard datasets can be used.

Dataset name	Domain	Description
SemEval-2014 (Task 4)	Laptops and restaurants	The ‘gold standard’ for ABSA research. Highly annotated.
AWARE (2025)	Smartphone apps	Modern dataset covering Productivity, Social, and Games.
M-ABSA (2025)	Multilingual	Covers 21 languages and 7 domains (travel, movies, etc).
Amazon/Yelp ABSA	E-commerce	Subsets of Amazon reviews specifically labelled for feature-level sentiment.